AWARE, Beyond Sentence Boundaries: A Contextual Transformer Framework for Identifying Cultural Capital in STEM Narratives

***Abstract*—Identifying cultural capital (CC) themes in student reflections can offer valuable insights that help foster equitable learning environments in classrooms. However, themes such as aspirational goals or family support are often woven into narratives, rather than appearing as direct keywords. This makes them difficult to detect for standard NLP models that process sentences in isolation. The core challenge stems from a lack of awareness, as standard models are pre-trained on general corpora, leaving them blind to the domain-specific language and narrative context inherent to the data. To address this, we introduce AWARE, a framework that systematically attempts to improve a transformer model's awareness for this nuanced task. AWARE has three core components: 1) Domain Awareness, adapting the model's vocabulary to the linguistic style of student reflections; 2) Context Awareness, generating sentence embeddings that are aware of the full essay context; and 3) Class Overlap Awareness, employing a multi-label strategy to recognize the coexistence of themes in a single sentence. Our results show that by making the model explicitly aware of the properties of the input, AWARE outperforms a strong baseline by 2.1 percentage points in Macro-F1 and shows considerable improvements across all themes. This work provides a robust and generalizable methodology for any text classification task in which meaning depends on the context of the narrative.**

I. Introduction

Recognizing the assets students bring to the classroom is fundamental to creating equitable learning environments. This challenge is particularly acute in fields like STEM, where a culture focused on a narrow definition of merit often overlooks the diverse strengths rooted in students' backgrounds. The Community Cultural Wealth framework [1] provides a lens for identifying forms of Cultural Capital (CC) such as aspirational, familial, and social capital. Prior research has shown that these themes often surface in student reflections [2]. Identifying these themes is directly linked to improved student engagement and success; however, identification of CC Themes at scale still remains a significant challenge.

The manual analysis of student writing is a major bottleneck. It requires researchers to undergo months of training to interpret how these complex themes manifest, and even then, the process is slow and subject to inter-annotator disagreement. While machine learning methods offer a path forward, they face a fundamental obstacle: CC themes do not appear as direct keywords or isolated sentences but are embedded in the meaning of the narrative. This context-dependent nature, coupled with the limited size of labeled data, makes the task challenging for standard text classifiers.

Our analysis revealed three core properties of our data that render conventional sentence-level models insufficient:

- **Domain-Specific Language:** Student reflections contain a unique vocabulary and style not well-represented in the general-language corpora on which models are pre-trained.
- **Context Dependency:** The meaning of a sentence is often contingent on the surrounding narrative. A sentence like, "They helped me to..." is ambiguous without the full essay context.
- **Theme Overlap:** CC themes are not mutually exclusive, and a single sentence can embody multiple themes. For instance, a sentence about an older sibling's guidance may reflect both Familial and Social capital.

Prior work has largely relied on sentence-level classifiers that, by design, are blind to these properties. To bridge this gap, we propose the AWARE framework. Our central hypothesis is that by explicitly making a model aware of these inherent properties of the data, we can enable it to detect CC themes more accurately. Accordingly, this paper details the architecture of the AWARE framework and presents a rigorous experimental validation of our approach, demonstrating a more robust and generalizable methodology for understanding nuanced student narratives.

The remainder of this paper is organized as follows: Section II reviews related work, Section III describes the AWARE framework, Section IV outlines our experimental setup, and Section V presents and analyzes the results, followed by our conclusion.

II. Related Work

Our work stands at the intersection of natural language processing and educational research. The novelty of our work lies in a methodological synthesis, which includes a novel context-aware architecture designed for this task. We purposefully integrate this architecture with established NLP techniques to build a model that is holistically "aware" of our data's unique properties. The three areas of research below directly inform the components of the AWARE framework.

A. Domain-Adaptive Pretraining (DAPT)

To address the challenge of Domain-Specific Language, we turn to domain adaptation. At its core, a language model works by learning the probabilistic patterns of word sequences. However, large language models like DeBERTa [3] are pre-trained on vast, general-domain corpora, and their performance can degrade when applied to specialized domains due to domain shift. An effective remedy, proposed by Gururangan et al. [4], is to continue the pre-training process on an in-domain corpus. This step, Domain-Adaptive Pre-training (DAPT), fine-tunes the model's internal representations to the vocabulary and nuances of the target domain. We adopt DAPT to make our model aware of the linguistic style of student writing, creating a stronger foundation for classification.

B. Context-Aware Modeling: A Top-Down Approach

To solve the problem of Context Dependency, we draw inspiration from hierarchical text modeling but invert the traditional approach. Standard classifiers process sentences in isolation, ignoring the inter-sentence relationships critical for narrative understanding. While methods like the Hierarchical Attention Network [5] address this by building document representations from sentences in a *bottom-up* fashion, our framework employs a *top-down* philosophy. Instead of building up from sentences, our model first encodes the entire essay to create globally-aware token embeddings. From these, we derive sentence-level representations using an attention pooling mechanism. As shown by Reimers & Gurevych [6], pooling token embeddings produces a more semantically meaningful sentence representation than relying on a single `[CLS]` token. This sequence of context-rich sentence embeddings is then passed to a Bidirectional LSTM (BiLSTM) to model the narrative flow, ensuring each sentence's representation is aware of its place within the larger story.

C. Multi-Label Classification for Theme Overlap

To handle Theme Overlap, we frame our task as multi-label classification. Simpler binary decomposition methods are ill-suited for this problem's structure. A one-vs-rest approach, where a classifier is trained for each theme against all other themes plus untagged sentences, is particularly problematic. The "rest" category forms a high-variance, poorly-defined negative class; it combines true negatives (no theme) with other, unrelated positive classes. This heterogeneity makes it difficult for a model to learn a clean, effective decision boundary, as the semantic distance between the target class and some members of the "rest" class can be smaller than the distance between members within the "rest" class itself.

Given that a single sentence can embody multiple forms of Cultural Capital, a standard single-label (softmax-based) approach would also fundamentally misrepresent the data by forcing a single, mutually exclusive choice. By treating each theme as an independent binary classification task (a binary relevance method) and using a sigmoid activation for each output neuron, we allow the model to predict a vector of probabilities. This enables the model to become aware of theme co-occurrence patterns, thereby capturing the complexity of the original narratives without creating artificial and noisy negative classes.

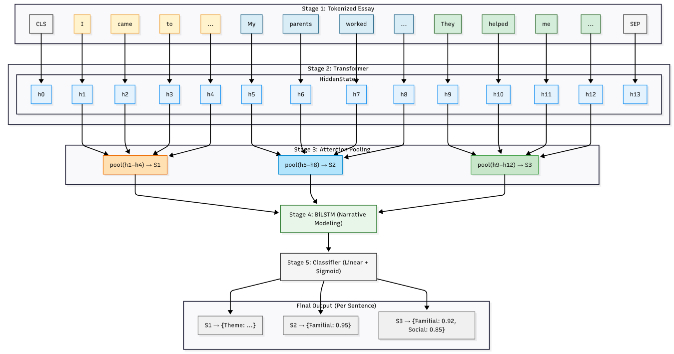

Fig. 1. The architecture of the AWARE framework. An essay is tokenized and processed by a transformer. Attention pooling is used to create sentence embeddings from the transformer's hidden states. A BiLSTM then models the narrative sequence of these sentences before a final classifier produces multi-label predictions for each sentence.

III. The AWARE Framework

The AWARE framework is a deep learning model designed to be context-aware. As illustrated in Fig. 1, the framework processes entire essays to generate contextually informed, sentence-level predictions. We build our framework upon a powerful pre-trained transformer, microsoft/deberta-v3-large, and augment it with components that systematically provide the model with the three critical forms of awareness discussed previously.

A. Domain Awareness via DAPT

Before being used in the main architecture, the DeBERTa model is first made Domain-Aware. State-of-the-art models like DeBERTa are trained on broad, general-purpose text. This generalization becomes a limitation when applied to a specialized domain like student reflections. To overcome this domain shift, AWARE's first step is Domain-Adaptive Pretraining (DAPT). This is an unsupervised learning step where we continue the pre-training of the DeBERTa model on our entire corpus of 1,499 student essays. Crucially, the model is not shown any class labels and learns only from the raw, unlabeled text using a masked language modeling (MLM) objective. During MLM, 15% of the input tokens are randomly masked, and the model is trained to predict these original tokens based on the surrounding context. This step fine-tunes the model's internal representations to the specific "dialect" of our target domain, providing a stronger foundation for the downstream task.

B. Context Awareness

The core of our framework is an architecture designed to be Context-Aware. Standard sentence-level classification fails because it loses the essay's narrative flow when it segments the text into sentences and often relies on a suboptimal sentence representation (the [CLS] token). Our architecture addresses these issues in a top-down manner across several stages:

Essay Reconstruction & Encoding: In a pre-processing step, sentences are grouped using their essay id to reconstruct the original essays. The full text of each essay is then fed into the DAPT-tuned DeBERTa encoder. This step produces contextualized token embeddings where the representation of each token is influenced by every other token in the entire essay. To keep track of which tokens belong to which sentence, we use the character offsets of each original sentence to identify its exact token span within the full essay.

Attention Pooling for Sentence Embeddings: To move from a token-level to a sentence-level understanding, we use the mappings from the previous step to isolate the token embeddings for each sentence. We then apply an attention pooling layer to these token groups. For each sentence, we compute a weighted average of its corresponding token embeddings. As shown by Reimers & Gurevych [5], this creates a richer semantic representation than relying on a single [CLS] token. The output is a sequence of sentence embeddings, S1,S2,..., Sn, as shown in Fig. 1.

Sequential Modeling: This sequence of sentence embeddings is then passed to a Bidirectional Long Short-Term Memory (BiLSTM) network. We chose a BiLSTM because it is exceptionally well-suited for modeling narrative. The BiLSTM reads the sequence of sentences in two directions: a forward pass reads from start to finish, gathering context from what came before, while a backward pass reads from finish to start, gathering context from what comes after. By combining these two perspectives, the model creates a final representation for each sentence that is fully aware of its role within the complete story. This is critical for resolving ambiguities, such as identifying who "they" refers to in "They helped me..." based on earlier sentences.

C. Class Overlap Awareness via Multi-Label Classification

The final component of the framework is designed to be Class Overlap-Aware. Simpler binary classification setups are insufficient because they cannot handle the inherent co-occurrence of CC themes. To address this, we formulate the task as a multi-label problem.

The contextually-enriched output from the BiLSTM for each sentence is passed through a final linear layer to produce a vector of logits, one for each potential CC theme. These logits are then passed through a sigmoid activation function, which outputs an independent probability for each theme. This enables the model to simultaneously predict multiple co-occurring themes within a single sentence. To train the model, we use a Focal Loss function [6], a modification of cross-entropy loss that is particularly effective for datasets with significant class imbalance, as it focuses training on harder-to-classify examples.

IV. Experimental Setup

To rigorously evaluate our framework, we designed an experimental setup that allows for a fair comparison between models and isolates the performance gains from each component of the AWARE pipeline. This includes a detailed data preparation pipeline, a clear ablation study, and a robust training and evaluation procedure.

A. Dataset and Pre-processing

The dataset originates from a research initiative at San Francisco State University [7], where student reflections on their experiences in STEM courses were collected over several semesters. Prompts such as "Why are you here?" were integrated into class activities, encouraging students to explore their motivations and lived experiences. The raw data presented several challenges for computational modeling, including inconsistent formatting, annotations as long excerpts rather than discrete sentences, and significant ambiguity in labeling.

To create a machine-learning-ready dataset, we performed a substantial restructuring process. We compiled a master list of all unique sentences from the corpus, linking each to its original essay ID to resolve duplication issues. We then created a multi-label binary vector for each sentence, indicating the presence (1) or absence (0) of seven Cultural Capital themes: Aspirational, Attainment, Navigational, Filial Piety, Familial, Social, and Spiritual. A special `class_0` label was used to explicitly indicate the absence of any theme.

The final dataset consists of 1,499 essays and 10,921 unique sentences. As shown in Table I, the class distribution is highly imbalanced.

TABLE I
CLASS DISTRIBUTION OF CULTURAL CAPITAL THEMES.

Theme	Sentence Count	Percentage of Total (10,921)
Aspirational	1,157	10.6%
Attainment	795	7.3%
Navigational	473	4.3%
Filial Piety	305	2.8%
Familial	269	2.5%
Social	226	2.1%
Spiritual	193	1.8%

Even the most frequent theme, Aspirational, is itself relatively rare, appearing in just over 10% of sentences, while the rarest theme, Spiritual, appears in a mere 1.8%. This significant imbalance poses a major challenge for classification and was a key motivation for employing techniques like Focal Loss and balanced sampling in our training procedure. Furthermore, our analysis confirms the necessity of a multi-label approach: of the 2,760 sentences containing at least one theme, 556 sentences (20.1%) contain multiple themes.

As shown in Fig. 2, the Jaccard similarity between themes reveals important relationships. For instance, Familial and Filial Piety have a high similarity score (0.49), indicating they frequently co-occur. This data-driven evidence underscores that a single-label classification approach would fail to capture the multifaceted nature of the student narratives, validating our choice of a multi-label framework.

B. Models for Ablation Study

To isolate the contribution of each component of our framework, we evaluated four variants of the model. Each model was

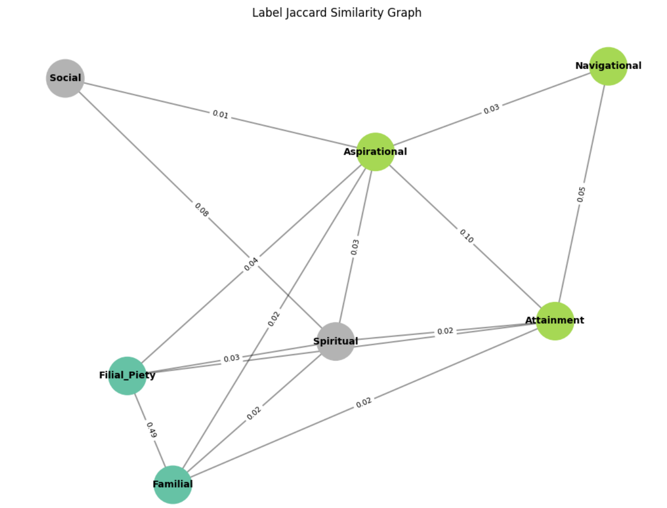

Fig. 2. Jaccard similarity between theme labels, illustrating the frequency of co-occurrence. The strong link between Familial and Filial Piety (0.49) highlights significant thematic overlap.

individually tuned using hyperparameter optimization (HPO) using the same resources to ensure a fair comparison.

- **Base Model:** A standard `microsoft/deberta-v3-large` classifier that processes each sentence independently with a maximum sequence length of 512 and a batch size of 32. During HPO, we tuned the encoder and decoder learning rates, weight decay, focal loss gamma, and label smoothing factor.
- **DAPT Model:** Identical to the Base Model, but initialized with weights from our domain-adaptive pre-training. This isolates the effect of Domain Awareness. It underwent the same HPO process as the Base Model.
- **Essay-Aware Model:** Uses the top-down, context-aware architecture (DeBERTa-BiLSTM with attention pooling) but without DAPT. To accommodate full essays, we increased the maximum sequence length to 1024. To maintain a comparable computational load to the Base Model, we made a corresponding trade-off by reducing the batch size to 4. The HPO search for this model was expanded to include parameters for the BiLSTM, such as dropout and hidden dimensions.
- **AWARE Model:** The full proposed model, combining the DAPT-tuned encoder with the context-aware architecture. To adhere to our computational constraints, this final model did not undergo its own separate, end-to-end HPO search. Instead, it was pragmatically constructed using the optimal components found in the prior tuning stages. It integrates the best-performing DAPT encoder with the optimal BiLSTM and classifier hyperparameters discovered during the HPO for the Essay-Aware model. The ”Post-HPO” results for this model represent the performance of this final, composed configuration.

This ablation study allows us to quantify the performance gains from each of our proposed “awareness” components.

C. Training and Hyperparameter Optimization

All experiments were conducted on the HPC cluster at San Francisco State University, using a single NVIDIA A100 GPU. A key real-world constraint was a total runtime limit of 8 hours for the entire training and evaluation pipeline for each model variant. This constraint informed our approach to HPO and other training decisions.

DAPT Implementation: The domain-adaptive pre-training was performed on the full corpus of 1,499 essays, split into a 90% training and 10% validation set. We used Optuna to tune key hyperparameters, and the final DAPT model was trained for up to 40 epochs with early stopping.

Cross-Validation and Data Splitting: For all downstream classification tasks, we used a 5-fold stratified cross-validation scheme at the essay level. This ensures that all sentences from one essay belong to the same fold, preventing data leakage. To guarantee that performance differences were due only to the model architectures, the exact same data folds were used for all models. For the Base Model, we took the sentences from the essay-level splits, dropped the essay identifiers, and shuffled the sentences, ensuring a direct sentence-to-sentence comparison.

Loss: We used the AdamW optimizer with a differential learning rate for the encoder and classifier layers. The learning rate was controlled by a linear scheduler with a 10% warmup period. We employed a Focal Loss function with label smoothing to handle class imbalance and a balanced sampling strategy during training, which began after the fifth epoch.

Hyperparameter Optimization (HPO): As noted above, we used Optuna to perform a systematic Bayesian Optimization search for key hyperparameters for each model variant, using its Tree-structured Parzen Estimator (TPE) sampler. This automated search was crucial for efficiently finding effective parameters for each specific architecture within our time constraints. The best-performing parameter set for each model was then used for the final cross-validation training. Detailed logs were maintained for each training run to monitor progress and performance.

D. Evaluation

Our primary evaluation metrics are Macro-F1 and Micro-F1 scores. Macro-F1 is particularly important as it gives equal weight to each theme, providing a better measure of performance on rare classes. We also report class-specific precision, recall, and F1 scores to analyze model behavior in detail. After training, we calibrate the decision threshold for each theme on the validation set to maximize F1-score.

All reported results are the mean across the 5 test folds, and the final predictions are generated using an ensemble of the five models trained during cross-validation.

V. Results and Analysis

We present a comprehensive analysis of the performance of our models, structured to first demonstrate the value of our foundational Domain-Adaptive Pretraining (DAPT), then to detail the performance of our classification models, and finally to analyze the remaining sources of error.

A. Analysis of Domain Adaptation (DAPT)

The effectiveness of DAPT is evident both quantitatively and qualitatively. As noted in Section III-A, the continued pretraining process dramatically reduced the model's perplexity on our domain-specific corpus from over 750,000 to 5.64. The practical impact of this adaptation is illustrated in Table II, which shows how the base and adapted models predict a masked word in a sentence characteristic of our dataset.

TABLE II
COMPARISON OF TOKEN PREDICTIONS FOR A MASKED WORD, SHOWING THE EFFECT OF DAPT.

Model	Input Text	Top Predicted Tokens
Original Model	"As a first-generation student, I had to learn how to navigate the university's [MASK] all by myself."	cuticle, biomedical, superstar
Adapted Model	"As a first-generation student, I had to learn how to navigate the university's [MASK] all by myself."	challenges, resources, process

The original model's predictions are random, whereas the adapted model predicts highly relevant, domain-specific words. This confirms that DAPT successfully tunes the model to the "dialect" of student writing, creating a strong foundation for classification. The quantitative improvements from this step are observed in the downstream classification task.

B. Classification Performance

We analyze the classification models in two stages: first, by evaluating the raw architectural impact before tuning, and second, by comparing the final optimized models after a rigorous Hyperparameter Optimization (HPO) process.

1) Architectural Impact Before HPO: Table III shows the per-theme F1-scores for each model variant before any tuning. This comparison isolates the raw performance gains attributable to our architectural choices.

TABLE III
PER-THEME F1-SCORE COMPARISON ACROSS MODELS (PRE-HPO).

Theme	Base	+DAPT only	+Context Aware only	AWARE	Improv.
Aspirational	0.86	0.89	0.87	0.89	+0.03
Attainment	0.54	0.55	0.55	0.57	+0.03
Familial	0.62	0.67	0.75	0.75	+0.13
Filial Piety	0.81	0.83	0.80	0.83	+0.02
Navigational	0.36	0.36	0.45	0.38	+0.02
Social	0.29	0.35	0.34	0.39	+0.10
Spiritual	0.62	0.67	0.62	0.65	+0.03
class_0	0.92	0.92	0.93	0.93	+0.01
Macro Avg	0.63	0.65	0.66	0.67	+0.04
Micro Avg	0.83	0.84	0.86	0.86	+0.03

Each component of the AWARE framework provides a clear benefit. The power of the context-aware architecture is particularly evident for highly contextual themes; for instance, the F1-score for *Familial* jumps by 13 points and *Navigational* by 9 points when comparing the Base and Essay-Aware models.

2) Final Tuned Model Comparison: After applying HPO to all variants—within a constrained 8-hour computational budget per model—we established our final, most robust results. Table IV provides a comprehensive summary of performance before and after tuning.

TABLE IV
OVERALL PERFORMANCE SUMMARY BEFORE AND AFTER HPO.

Model Variant	Pre-HPO		Post-HPO	
	Macro-F1	Micro-F1	Macro-F1	Micro-F1
Base Model	0.63	0.83	0.66	0.85
+DAPT only	0.65	0.84	0.65	0.85
+Context-Aware only	0.66	0.86	0.67	0.86
AWARE (DAPT + Context-Aware)	0.67	0.86	**0.68**	**0.86**

Systematic tuning improves all models, but the AWARE framework's superior architecture is the decisive factor. It is particularly noteworthy that the untuned AWARE model (Macro-F1: 0.672) already outperforms the final, heavily tuned Base Model (Macro-F1: 0.662). This highlights that the architectural design is the primary driver of performance, not merely the optimization process.

The final tuned AWARE model (Macro-F1: 0.683) ultimately outperforms the tuned Base Model by 2.1 percentage points. The AWARE model's gains are most significant in themes requiring nuanced understanding. For *Filial Piety* (+0.11 F1), the model dramatically boosts precision from 0.71 to 0.88 without sacrificing recall, indicating it learns a more accurate and less ambiguous representation of the theme. Similarly, for *Social* capital (+0.11 F1), the context-aware architecture substantially improves precision, leading to a more reliable classifier.

C. Error Analysis

Despite the AWARE framework's strong performance, an analysis of its predictions reveals key areas for future work. The detailed classification report for our best model shows two primary challenges:

Inherently Ambiguous Themes: The themes *Navigational* (F1: 0.40) and *Social* (F1: 0.47) consistently score the lowest across all models. While AWARE improves the F1-score for *Social* capital by 11 points over the optimized baseline, the overall score remains low. This suggests these concepts are either more semantically ambiguous in student writing or are underrepresented in the dataset, making them difficult for any model to learn robustly.

Precision/Recall Trade-off: For several themes, the model achieves high recall at the cost of lower precision. For example, in the final AWARE model, *Familial* capital has a recall of 0.86 but precision of 0.60. This indicates that while the model is effective at identifying most sentences with this theme, it also produces a notable number of false positives. This behavior is likely a deliberate trade-off influenced by our use of a class-balanced sampler, which encourages the model to predict minority classes more frequently.

These findings suggest that future efforts should focus on gathering more targeted data for ambiguous themes and exploring more advanced techniques to manage the precision-recall trade-off in imbalanced datasets.

VI. Conclusion

In this paper, we addressed the challenging task of identifying Cultural Capital themes in student narratives. We argued that standard classifiers are insufficient for this task due to the contextual, domain-specific, and overlapping nature of the themes. Our central contribution is the AWARE framework, which is built on the principle that explicitly making a model aware of a problem's core properties is the key to improving performance.

Our experiments validate this principle. We demonstrated that the AWARE architecture itself is the primary driver of success, with the untuned framework outperforming a fully optimized baseline. While our final model significantly improves performance, particularly on nuanced themes like Filial Piety and Social capital, the remaining challenges with the most ambiguous themes underscore the path still ahead.

The phrase "beyond sentence boundaries" thus carries a dual meaning for our work. Technically, it describes how our AWARE model operates: by processing entire essays to understand the narrative context that gives individual sentences their meaning. Methodologically, it reflects the social goal of our work: to look beyond the narrow boundaries of traditional merit and recognize the diverse cultural assets students bring into STEM classrooms. By enabling a more complete view of student stories, this approach fosters more culturally responsive teaching practices and provides a robust methodology for any domain where understanding the full story is paramount.

VII. Future Work

Our work establishes a strong foundation for context-aware analysis. We propose several promising directions for future research:

- **Addressing Ambiguity and Imbalance:** Our error analysis revealed that themes like *Navigational* and *Social* capital remain challenging, likely due to their low representation in the dataset. We plan to collect targeted data for these specific themes and explore more sophisticated methods for managing the precision-recall trade-off to create a higher-precision tool for educators.
- **Enhancing Awareness with External Knowledge:** We will extend the AWARE framework by incorporating new dimensions of awareness. This includes providing the model with explicit definitions of each Cultural Capital theme, leveraging knowledge graphs to encode the relationships between concepts, and exploring *Temporal Awareness* to track how a student's perspective evolves throughout a narrative.
- **Developing Actionable, Explainable Tools:** We aim to distill our model into an efficient version that can power real-time tools for educators. A crucial component of this will be building an interactive system that not only highlights themes but also provides supporting evidence from the text. Making the model's reasoning transparent is essential for building trust and positioning this technology as a collaborative partner for educators, not a black box.

VIII. Impact Statement

The immediate goal of this work is to provide a more accurate and efficient tool for educational researchers to identify Cultural Capital. By moving beyond simple keyword matching and respecting the narrative context of student stories, this tool can help educators recognize and validate the diverse assets students bring to STEM classrooms, fostering a greater sense of belonging and engagement. This supports a critical shift towards more equitable, asset-based pedagogies.

The broader impact of our methodology, however, extends beyond this specific application. We demonstrate that for any high-stakes domain where meaning is derived from narrative—such as analyzing patient histories in healthcare, legal documents, or social service records—standard sentence-level models risk not just misinterpretation, but flawed conclusions with serious consequences. Overlooking critical context can lead to overlooking a patient's full history, a key legal precedent, or a student's unique strengths.

Our work therefore serves as a case study for the critical importance of context-aware modeling. By advocating for systems that respect narrative integrity, we aim to contribute to a future where machine learning applications are more robust, accurate, and ultimately, more equitable.

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APPENDIX

A. Optimal Hyperparameters

The models in our ablation study were tuned using Optuna. Table V details the final, optimal hyperparameters discovered for each model variant. The full AWARE model represents the `dapt_essay_aware_optimized` configuration, which achieved the highest performance. All models used a random seed of 42 for reproducibility.

TABLE V
OPTIMAL HYPERPARAMETERS FOR MAIN MODEL VARIANTS.

Hyperparameter	Base	+DAPT	Essay-Aware	AWARE
General Parameters				
Encoder Learning Rate	4.88e-5	4.88e-5	3.69e-5	3.69e-5
Decoder Learning Rate	6.74e-5	6.74e-5	1.86e-4	1.86e-4
Weight Decay	0.00366	0.00366	0.00107	0.00107
Focal Loss Gamma (γ)	3.06	3.06	2.76	2.76
Label Smoothing	0.196	0.196	0.168	0.168
Architectural Parameters				
BiLSTM Hidden Dimensions	N/A	N/A	256	256
BiLSTM Dropout	N/A	N/A	0.212	0.212

B. Detailed Classification Reports

The following tables provide the detailed per-theme Precision, Recall, and F1-scores on the test set for the final, tuned versions of each model in the ablation study.

TABLE VI
CLASSIFICATION REPORT FOR TUNED **BASE MODEL**.

Theme	Precision	Recall	F1-Score
Aspirational	0.89	0.94	0.91
Attainment	0.41	0.81	0.54
Familial	0.53	0.93	0.68
Filial Piety	0.71	0.84	0.77
Navigational	0.31	0.49	0.38
Social	0.47	0.29	0.36
Spiritual	0.67	0.80	0.73

TABLE VII
CLASSIFICATION REPORT FOR TUNED **+DAPT MODEL**.

Theme	Precision	Recall	F1-Score
Aspirational	0.86	0.86	0.86
Attainment	0.41	0.73	0.52
Familial	0.54	0.93	0.68
Filial Piety	0.69	0.84	0.76
Navigational	0.32	0.51	0.39
Social	0.57	0.33	0.42
Spiritual	0.67	0.60	0.63

TABLE VIII
CLASSIFICATION REPORT FOR TUNED **ESSAY-AWARE MODEL**.

Theme	Precision	Recall	F1-Score
Aspirational	0.91	0.90	0.91
Attainment	0.46	0.69	0.55
Familial	0.51	1.00	0.67
Filial Piety	0.85	0.72	0.78
Navigational	0.46	0.47	0.46
Social	0.58	0.29	0.39
Spiritual	0.79	0.55	0.65

TABLE IX
CLASSIFICATION REPORT FOR FINAL TUNED **AWARE MODEL**.

Theme	Precision	Recall	F1-Score
Aspirational	0.89	0.87	0.88
Attainment	0.44	0.64	0.52
Familial	0.60	0.86	0.71
Filial Piety	0.88	0.88	0.88
Navigational	0.45	0.37	0.40
Social	0.64	0.38	0.47
Spiritual	0.72	0.65	0.68